

Abdellah Fourtassi

Associate Professor (Maître de Conférences HDR) in Computer Science

Aix-Marseille Université · LIS / TALEP · Institute of Language, Communication & the Brain (ILCB)
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RESEARCH FOCUS I build machine-learning models that learn from multimodal interaction under cognitively plausible constraints. The goal is to develop formal theories of how interaction shapes language learning in humans and machines.

ACADEMIC APPOINTMENTS & ROLES

Aix-Marseille Université - Associate Professor (Maître de Conférences HDR), Computer Science	2025-present
Aix-Marseille Université - Assistant Professor (Maître de Conférences), Computer Science	2019-2025
Director, Computation in Communicative Development (CoCoDev) team	2019-present
Deputy Director for Research, Institute of Language, Communication & the Brain (ILCB)	2024-present
Co-lead, <i>NLP & Cognition</i> national working group, CNRS GDR TAL	2025-present
Co-Director, Master in Cognitive Science (MASCO), Aix-Marseille Université	2020-2021
Associate Editor, <i>Dialogue & Discourse</i>	2026-present
Visiting Researcher, INRIA Paris (Délégation Recherche)	2021-2022
Postdoctoral Research Fellow, Department of Psychology, Stanford University	2016-2019

EDUCATION

PhD in Cognitive Science, École Normale Supérieure (ENS), Paris - summa cum laude	2015
Master's in Cognitive Science (CogMaster), EHESS, Paris	2012
Science & Engineering Diploma, École Polytechnique, Paris	2012
CPGE/MP*, Mathematics & Physics, Lycée Moulay Youssef, Rabat	2006-2008

RESEARCH FUNDING

ANR JCJC - Principal Investigator, €287k	2022-2025
<i>Mastering the Art of Conversation in Middle Childhood (MACoMiC)</i>	

SELECTED HONORS & DISTINCTIONS

- Keynote speaker: SIGDIAL (2025), LM Playschool Workshop at EMNLP (2026), Child Language Symposium (2026)
- Aix-Marseille Université Best PhD Award (2024), awarded to M. Nikolaus; co-supervised with L. Prévot
- Marie Skłodowska-Curie European Fellowship, €185k (2019; declined after accepting tenured position)
- Stanford Data Science Seed Grant, \$50k (2018)
- Fyssen Foundation Postdoctoral Study Grant, €40k (2016)
- France-Stanford Center Collaborative Project Award, \$15k (2015-2016)
- CoNLL Best Paper Award finalist (for Fourtassi & Dupoux, 2014)
- ACL Best Student Paper Award, \$1,000 (for Fourtassi & Dupoux, 2013)

PUBLICATIONS

Book

1. Fourtassi, A. (forthcoming). *Neural Networks for a Theory of Language Development*. Cambridge University Press.

Edited collections

2. Agee, L., Monfils, M., & Fourtassi, A. (2025). Editorial: The Neural and Behavioral Methods of Social Learning. *Frontiers in Human Neuroscience*.
3. Apidianaki, M., Fourtassi, A., & Padó, S. (2024). Editorial: Language Learning, Representation, and Processing in Humans and Machines. *Computational Linguistics*.

Journal articles

4. Brahimi, Y., & Fourtassi, A. (under review). Slow but Predictive: Turn-Taking Development in Child-Adult Conversation.
5. Agrawal, A., Favre, B., & Fourtassi, A. (in press). Scaffolding Early Dialogue: A Unified Account of Response Contingency in Child-Caregiver Interaction. *Cognitive Science*.
6. Nikolaus, M., & Fourtassi, A. (2026). Modeling Children's Grammar Learning via Caregiver Feedback in Natural Conversations. *Philosophical Transactions of the Royal Society B*, 381.

7. Gupta, S., Mazzocconi, C., & Fourtassi, A. (2025). Toward a child-centered, interactive approach to multimodal language development. *First Language*, 45(6).
8. Mazzocconi, C., O'Brien, B., Bodur, K., & Fourtassi, A. (2025). Do Children Laugh Like Their Parents? Conversational Laughter Mimicry and Acoustic Alignment in Middle Childhood. *Journal of Nonverbal Behavior*, 49.
9. Bodur, K., Fourtassi, A., Fredouille, C., & Meunier, C. (2025). Speech Reduction in Middle Childhood: The Case of Child-Caregiver Interactions. *First Language*.
10. Agee, L., Fourtassi, A., & Monfils, M. (2025). Social Context as a Source of Variability in the Psychological Sciences. *Frontiers in Behavioral Neuroscience*, 18.
11. Cabiddu, F., Nikolaus, M., & Fourtassi, A. (2025). Comparing Children and Large Language Models in Word Sense Disambiguation: Insights and Challenges. *Language Development Research*, 5(1).
12. Bodur, K., Nikolaus, M., Prévot, L., & Fourtassi, A. (2023). Using Video Calls to Study Children's Conversational Development: The Case of Backchannel Signaling. *Frontiers in Computer Science*, 5.
13. Wilson, K., Frank, M. C., & Fourtassi, A. (2023). Conceptual Hierarchy in Child-Directed Speech: Implicit Cues Are More Reliable. *Journal of Cognition & Development*, 24(4).
14. Nikolaus, M., & Fourtassi, A. (2023). Communicative Feedback in Language Acquisition. *New Ideas in Psychology*, 68, 100985.
15. Nikolaus, M., Maes, E., Auguste, J., Prévot, L., & Fourtassi, A. (2022). Large-Scale Study of Speech Acts' Development in Early Childhood. *Language Development Research*, 2(1).
16. Jiang, H., Frank, M. C., Kulkarni, V., & Fourtassi, A. (2022). Stability and Change in Caregivers' Word Usage Across Early Childhood. *Cognitive Science*, 46(7).
17. Fourtassi, A., Regan, S., & Frank, M. C. (2021). Continuous Developmental Change Explains Discontinuities in Word Learning. *Developmental Science*, 24(2).
18. Fourtassi, A., & Frank, M. C. (2020). How Optimal Is Word Identification Under Multimodal Uncertainty? *Cognition*, 199.
19. Fourtassi, A., Bian, Y., & Frank, M. C. (2020). The Growth of Children's Semantic and Phonological Networks: Insights from 10 Languages. *Cognitive Science*, 44(7).

Peer-reviewed conference proceedings

20. Jumelet, J., Fourtassi, A., Haga, A., Bunzeck, B., ... Choshen, L. (2026). BabyBabelLM: A Multilingual Benchmark of Developmentally Plausible Training Data. EACL (19th Conference of the European Chapter of the Association for Computational Linguistics).
21. Brahimi, Y., Blanc, C., & Fourtassi, A. (2026). Predicting Turn-Taking in Child-Adult Conversations Using Voice Activity Projection. IWSDS (16th International Workshop on Spoken Dialogue Systems Technology).
22. Liu, J., & Fourtassi, A. (2025). Benchmarking LLMs for Mimicking Child-Caregiver Language in Interaction. SemDial (29th Workshop on the Semantics and Pragmatics of Dialogue).
23. Agrawal, A., Favre, B., & Fourtassi, A. (2025). Identifying Repair Opportunities in Child-Caregiver Interactions. SemDial (29th Workshop on the Semantics and Pragmatics of Dialogue).
24. Agrawal, A., Nikolaus, M., Favre, B., & Fourtassi, A. (2024). Automatic Coding of Contingency in Child-Caregiver Conversations. LREC-COLING (The Joint International Conference on Computational Linguistics, Language Resources, and Evaluation).
25. Nikolaus, M., Agrawal, A., Kaklamanis, P., Warstadt, A., & Fourtassi, A. (2024). Automatic Annotation of Grammaticality in Child-Caregiver Conversations. LREC-COLING (The Joint International Conference on Computational Linguistics, Language Resources, and Evaluation).
26. Goumri, D., ..., & Fourtassi, A. (2024). A Developmental Corpus of Child-Caregiver Face-to-Face vs. Video-Call Conversations in Middle Childhood. LREC-COLING (The Joint International Conference on Computational Linguistics, Language Resources, and Evaluation).
27. Schweitzer, M., Montagnini, A., Fourtassi, A., & Schatz, T. (2024). Multimodal Integration in Audio-Visual Speech Recognition. NeurIPS Behavioral Machine Learning Workshop.
28. Goumri, D.-E., Becerra, L., & Fourtassi, A. (2024). Child-Caregiver Gaze Dynamics in Naturalistic Face-to-Face Conversations. CogSci (46th Annual Meeting of the Cognitive Science Society).
29. Agrawal, A., Favre, B., & Fourtassi, A. (2024). Analyzing Communicative Intent Coordination in Child-Caregiver Interactions. CogSci (46th Annual Meeting of the Cognitive Science Society).
30. Peirola, M., Xu, Z., & Fourtassi, A. (2024). Development of Flexible Role-Taking in Conversations Across Preschool. CogSci (46th Annual Meeting of the Cognitive Science Society).

31. Goumri, D., Janssoone, T., Becerra, L., & Fourtassi, A. (2023). Automatic Detection of Gaze and Smile in Children's Video Calls. *ICMI Companion (International Conference on Multimodal Interaction)*.
32. Cabiddu, F., Nikolaus, M., & Fourtassi, A. (2023). Modeling Children's Word Sense Disambiguation with Large Language Models. *CogSci (45th Annual Meeting of the Cognitive Science Society)*.
33. Nikolaus, M., Prévot, L., & Fourtassi, A. (2023). Communicative Feedback Signals in Response to Children's Grammatical Errors. *CogSci (45th Annual Meeting of the Cognitive Science Society)*.
34. Mazzocconi, C., O'Brien, B., El Haddad, K., Bodur, K., & Fourtassi, A. (2023). Mimicking and Non-Mimicking Laughter in Child-Caregiver Conversation. *CogSci (45th Annual Meeting of the Cognitive Science Society)*.
35. Agrawal, A., Liu, J., Bodur, K., Favre, B., & Fourtassi, A. (2023). Development of Multimodal Turn Coordination in Conversations. *CogSci (45th Annual Meeting of the Cognitive Science Society)*.
36. Tiran, T., Meewis, F., Fourtassi, A., & Dautriche, I. (2023). Typology of Topological Relations Using Machine Translation. *CogSci (45th Annual Meeting of the Cognitive Science Society)*.
37. Nikolaus, M., Salin, E., Ayache, S., Fourtassi, A., & Favre, B. (2022). Do Vision-and-Language Transformers Learn Predicate-Noun Dependencies? *EMNLP (Conference on Empirical Methods in Natural Language Processing)*.
38. Liu, J., Nikolaus, M., Bodur, K., & Fourtassi, A. (2022). Predicting Backchannel Signaling in Multimodal Child-Caregiver Conversation. *ICMI Companion (International Conference on Multimodal Interaction)*.
39. Misiek, T., & Fourtassi, A. (2022). Caregivers Exaggerate Their Lexical Alignment to Young Children Across Several Cultures. *SemDial (26th Workshop on the Semantics and Pragmatics of Dialogue)*.
40. Hallart, C., Peirolo, M., Xu, Z., & Fourtassi, A. (2022). Development of Dialogue Contingency in Child-Caregiver Interaction. *SemDial (26th Workshop on the Semantics and Pragmatics of Dialogue)*.
41. Bodur, K., Nikolaus, M., Prévot, L., & Fourtassi, A. (2022). Backchannel Behavior in Child-Caregiver Video Calls. *CogSci (44th Annual Meeting of the Cognitive Science Society)*.
42. Nikolaus, M., Prévot, L., & Fourtassi, A. (2022). Communicative Feedback Supports Intelligible Speech in Childhood. *CogSci (44th Annual Meeting of the Cognitive Science Society)*.
43. Nikolaus, M., & Fourtassi, A. (2021). Interaction Between Perception- and Production-Based Learning in Early Semantic Development. *CoNLL (25th Conference on Computational Natural Language Learning)*.
44. Dary, F., Nasr, A., & Fourtassi, A. (2021). TALEP at CMCL 2021 Shared Task: Predicting Eye-Tracking Data (ranked 2 out of 30 submissions). *CMCL (11th Workshop on Cognitive Modeling and Computational Linguistics)*.
45. Bodur, K., Nikolaus, M., Kassim, F., Prévot, L., & Fourtassi, A. (2021). ChiCo: A Multimodal Corpus for the Study of Child Conversation. *ICMI Companion (International Conference on Multimodal Interaction)*.
46. Nikolaus, M., & Fourtassi, A. (2021). Evaluating the Acquisition of Semantic Knowledge from Cross-situational Learning in Artificial Neural Networks. *CMCL (11th Workshop on Cognitive Modeling and Computational Linguistics)*.
47. Nikolaus, M., Maes, E., Auguste, J., Prévot, L., & Fourtassi, A. (2021). Large-scale Study of Speech Acts' Development Using Automatic Labeling. *CogSci (43rd Annual Meeting of the Cognitive Science Society)*.
48. Nikolaus, M., Maes, E., & Fourtassi, A. (2021). Modeling Speech Act Development in Early Childhood: The Role of Frequency and Linguistic Cues. *CogSci (43rd Annual Meeting of the Cognitive Science Society)*.
49. Dary, F., Fourtassi, A., & Nasr, A. (2021). On the Role of Low-level Linguistic Tasks for Reading Time Prediction. *CogSci (43rd Annual Meeting of the Cognitive Science Society)*.
50. Misiek, T., Favre, B., & Fourtassi, A. (2020). Development of Multi-level Linguistic Alignment in Child-Adult Conversations. *CMCL (10th Workshop on Cognitive Modeling and Computational Linguistics)*.
51. Fourtassi, A. (2020). Word Co-Occurrence in Child-Directed Speech Predicts Children's Free Word Associations. *CMCL (10th Workshop on Cognitive Modeling and Computational Linguistics)*.
52. Fourtassi, A., Wilson, K., & Frank, M. C. (2020). Discovering Conceptual Hierarchy Through Explicit and Implicit Cues. *CogSci (42nd Annual Meeting of the Cognitive Science Society)*.
53. Fourtassi, A., Scheinfeld, I., & Frank, M. C. (2019). The Development of Abstract Concepts in the Children's Early Lexical Networks. *CMCL (10th Workshop on Cognitive Modeling and Computational Linguistics)*.
54. Fourtassi, A., Regan, S., & Frank, M. C. (2019). The Role of Continuous Developmental Change in Explaining Discontinuities in Word Learning. *CogSci (41st Annual Meeting of the Cognitive Science Society)*.
55. Fourtassi, A., & Dupoux, E. (2019). Phoneme Learning is Influenced by the Taxonomic Similarity of the Semantic Referents. *CogSci (41st Annual Meeting of the Cognitive Science Society)*.
56. Fourtassi, A., Bian, Y., & Frank, M. C. (2018). Word Learning as Network Growth: A Cross-Linguistic Analysis. *CogSci (40th Annual Meeting of the Cognitive Science Society)*.

57. Fourtassi, A., & Frank, M. C. (2017). Word Identification Under Multimodal Uncertainty. CogSci (39th Annual Meeting of the Cognitive Science Society).
58. Fourtassi, A., & Dupoux, E. (2016). The Role of Word-Word Co-Occurrence in Word-Meaning Learning. CogSci (38th Annual Meeting of the Cognitive Science Society).
59. Fourtassi, A., & Dupoux, E. (2014). Rudimentary Lexicon and Semantics Help Bootstrap Phoneme Acquisition. CoNLL (18th Conference on Computational Natural Language Learning).
60. Fourtassi, A., Schatz, T., Varadarajan, B., & Dupoux, E. (2014). Exploring the Relative Role of Bottom-Up and Top-Down Information in Phoneme Learning. ACL (52nd Annual Meeting of the Association for Computational Linguistics).
61. Fourtassi, A., Dunbar, E., & Dupoux, E. (2014). Self-Consistency as an Inductive Bias in Early Language Acquisition. CogSci (36th Annual Meeting of the Cognitive Science Society).
62. Fourtassi, A., Borschinger, B., Johnson, M., & Dupoux, E. (2013). Why Is English So Easy to Segment? CMCL (The 4th Workshop on Cognitive Modeling and Computational Linguistics).
63. Fourtassi, A., & Dupoux, E. (2013). A Corpus-Based Evaluation Method for Distributional Semantic Models. ACL-SRW (51st Annual Meeting of the Association for Computational Linguistics).
64. Jansen, A., et al. (2013). Zero-Resource Speech Technologies and Models of Early Language Acquisition. ICASSP (The International Conference on Acoustics, Speech and Signal Processing).

INVITED TALKS

Keynotes

- LM Playschool Workshop at EMNLP (Budapest, 2026)
- Child Language Symposium (London, 2026)
- SIGDIAL (Avignon, 2025)

Selected invited lectures and expert meetings

- University of Vienna, Current topics in AI, guest lecture (2026)
- International Congress of Infant Studies (ICIS, online event, 2025)
- Mila Quebec AI Institute workshop on NLP & Cognitive Science (2024)
- UCL, Psychology and Language Sciences (2023, 2024)
- Dagstuhl Seminar: Developmental Machine Learning (2022)
- OECD workshop: AI and the Productivity of Science (2021)

MENTORSHIP

PhD supervision - primary advisor

- Marianne Schweitzer (2025-present), co-advised with C. François
- Youcef Brahimi (2025-present)
- Dhia Elhak Goumri (2022-2026), co-advised with L. Becerra; defended March 2026
- Abhishek Agrawal (2022-2025), co-advised with B. Favre; defended October 2025
- Mitja Nikolaus (2020-2023), co-advised with L. Prévot; defended October 2023; *AMU Best Ph.D. Award 2024*

PhD supervision - secondary advisor

- Mathias Vigouroux (2025-present), Jing Liu (2024-present), Franck Dary (2018-2022)

Postdoctoral supervision

- Giulia Rambelli (2026-present), Monica Barbir (2024-present), Chiara Mazzocchi (2020-2023)

Additional supervision

- Supervised 14 Master's interns, 17 summer interns; multiple projects resulted in peer-reviewed publications
- Member of 13 thesis advisory committees

TEACHING

- Building Minds: The Programming Tools - Master in Cognitive Science (2025-present)
- Machine Learning for Cognitive Science - MASCO / Master in Computer Science (2024-present; with T. Schatz)
- Data Science (L1 Computer Science, 2025-present)
- Introduction to Machine Learning (L3 Computer Science, 2022-2023)
- Introduction to Natural Language Processing (M1 Computer Science, 2020-2023)
- Probability Theory for Computer Science (M1, 2022)

- Theories in Cognitive Science (MASCO, 2020-2022)
- ILCB Summer School: week-long course on deep learning and language development (2025, 2026)
- ETAL National Summer School on NLP: When NLP Meets Cognitive Science (2023)

EDITORIAL, REVIEWING & SERVICE

Editorial and conference roles

- Associate Editor, *Dialogue & Discourse* (2026-present)
- Area Chair, ACL Rolling Review (2025-present)
- Program Chair, ACL Workshop on Computational Developmental Linguistics (2026)
- Meta-Reviewer, CogSci (2026)
- Senior Area Chair, COLING (2025)
- Associate Editor, *Frontiers in Cognitive Neuroscience* (2023-2025)
- Guest Editor, *Computational Linguistics* (2024) and *Frontiers in Human Neuroscience* (2025)

Reviewing

- Funding agencies: ANR, NSF, British Academy, Royal Society, JST
- Selected journals: Psychological Review, Trends in Cognitive Sciences, Proceedings B, Cognition, Cognitive Science, Child Development, Developmental Psychology, JEP: General, Behavior Research Methods
- Selected conferences: ACL Rolling Review, CogSci, ICMI, CoNLL, CDS, ACII, CMCL

Service & Events (recent)

- Co-organizer, national workshop on *Developmental AI*, CNRS GDR TAL / Babylab, 2026 (with M. Fort, T. Hueber)
- Evaluation panel, France-Japan bilateral funding, ANR-JST CREST, 2026
- Co-organizer, CIVIS Europe-wide collaborative workshop, 2026 (with M. Jover)
- Co-organizer, national workshop on *NLP & Cognition*, CNRS GDR TAL, 2025 (with P. Denis, L. Prévot)
- Selection panel, Associate Professor position in Computer Science, Université de Lille, 2025

PUBLIC OUTREACH

- Bringing the lab to the classroom, ANR-funded Festival Explore (2026)
- Interview: AI and Cognitive Science, Photizo podcast (2025)
- Interactive MACoMiC project stand, Nuit des chercheurs (2023)
- Public outreach article in *The Conversation* on infants learning conversation (2022)